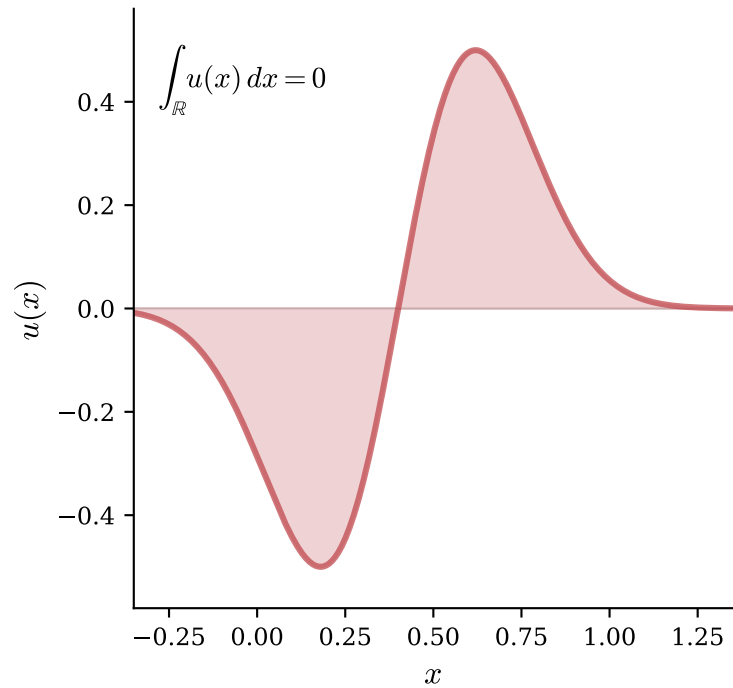


Density variation, cumulative distribution, and quantile

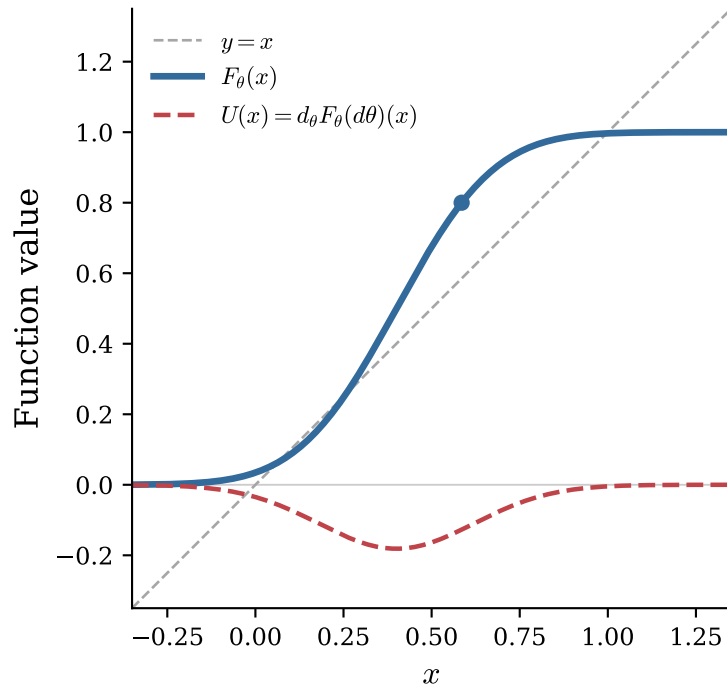
$$p_\theta = \mathcal{N}(\mu, \sigma^2), \quad \theta = (0.40, 0.22), \quad d\theta = (0.10, 0)$$

(a) $u = d_\theta p_\theta(d\theta)$



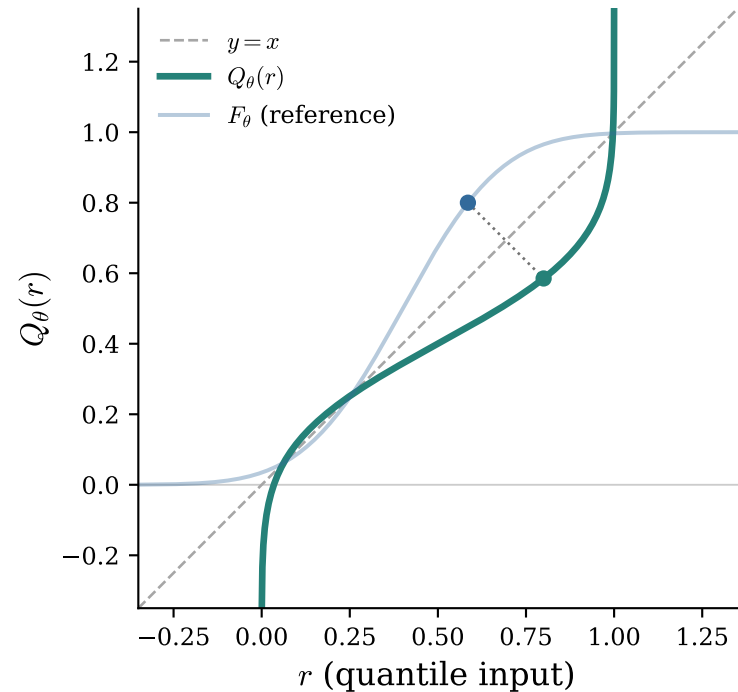
$$u(x) = d\mu \frac{x - \mu}{\sigma^2} p_\theta(x)$$

(b) F_θ and its variation U



$$F_\theta(x) = \Phi((x - \mu)/\sigma), \quad U(x) = -d\mu p_\theta(x)$$

(c) $Q_\theta = F_\theta^{-1}$



$$Q_\theta(r) = \mu + \sigma \Phi^{-1}(r), \quad 0 < r < 1$$